

OFFLINE: train and prepare

Learn from training snapshots

Spatial bank model,
head h_θ , training codes

Prepare the selected configuration

Bank G , corrections C_q , weak operators

Frozen during the online solve

ONLINE

PDE inputs

Source or initial
field; parameters

Initialize

Initial reduced
coordinates
(z_0, y_0)

Reduced solve

Weak residual
minimization
Adjust z and y

Reconstruct

$u = G[h_\theta(z) + C_q y]$
Requested
solution fields

Residual evaluation:

dense or optional EQ

Empirical quadrature uses
stored points and weights

q : number of correction directions.
Tolerance and iteration cap control
the solve.

G : frozen spatial bank; h_θ : frozen neural head; C_q : fixed correction directions; z : nonlinear coordinates; y : correction coefficients. Solid arrows show online flow or an evaluation choice; dashed arrows show reuse of prepared components.